

FACULTY OF ENGINEERING AND GREEN TECHNOLOGY

UGEA3183 DIGITAL SIGNAL PROCESSING

LABORATORY REPORT 2

Prepared By:

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(*Lab Report Evaluation*)

Course: UGEA2183 Digital Signal Processing
Lab 1 and Lab 2

No	Criteria	10 Excellent	8 Good	5 Fair	2 Poor	Score
1	Presentation of **results**	Results are clearly stated and presented (in table or graphics if necessary).	Results are stated and presented (in table or graphics if necessary).	Results are fairly stated and presented (in table or graphics if necessary).	The results are wrong	*X 1.0*
2	Analysis and **discussion** of the findings and	Neatly tabulated and presented; with complete and accurate results. Able to elaborate, interpret and evaluate the results analytically.	Neatly tabulated and presented; accurate result but missing some minor data. Able to elaborate, interpret and evaluate result analytically; but minor improvements are needed.	Neatly tabulated and presented; but an incomplete and inaccurate result. Able to elaborate, interpret and evaluate results analytically, but missing some important points.	Major data are missing. Unable to elaborate, interpret, and evaluate the result.	*X1.0*
	Conclusion	Summarizes all important points; appropriately addressed the objective, result and discussion.	All important points are summarized; missing some minor points, sufficient elaboration.	Not all important points are summarized; sufficient elaboration.	Concluding paragraph is not apparent.	
3	**Overall presentation** Report format, spelling, grammar.	Report is Organized. Spelling and grammar are correct.	Report is organized .< 5% errors in spelling, grammar.	Report is organized. 5-10% errors in spelling, grammar.	Report is unorganized. Very frequent errors in spelling, grammar..	*X 1.0*
4	**Ability to perform an experiment successfully without supervision**	Able to perform all experiments without assistance and supervision successfully and safely.	Able to perform 80% of the experiments without assistance and supervision successfully and safely.	Able to perform 50% of the experiments without assistance successfully and safely. Able to experiment safely under supervision.	Not able to experiment without assistance and supervision.	*X 1.0*
5	**Originality References** and citations are present. The report is not a copy of peers	Adequate ref and citation, report is original	Adequate but have <5% error in ref/citations.	Only a few ref/citations	No ref/citations at all	*X 1.0*
					Total mark	**/50**

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Introduction

FIR filter, is an acronym for Finite impulse filter, refer to a type of filter that uses frequency selection that uses only the input signal to determine the current output of the filter. The specification of the filter is that the output becomes stable eventually instead of oscillating infinitely. This filtering is done by convolution of an input signal in respect of another input impulse signal. FIR is emphasized in daily circuit usage due to the high reliability and stability.

In this experiment, the filter is designed using the MATLAB filter design tool, The sample is processed and multiply it by a combination of coefficient which is then added together to form and perform a FIR Filter.

The nature of FIR emphasizes that it does not have the feedback system, this will limit the possibility of indefinite oscillation. The FIR filter will only respond to a small, limited number of samples when a short impulse is given. After these samples, the response will approximate to zero (dspGuru, n.d.).

Results

Section 1: Low-Pass Filter design with F_s=8kHz, 10th Order, F_{pass}=2kHz, F_{stop}=2.25kHz

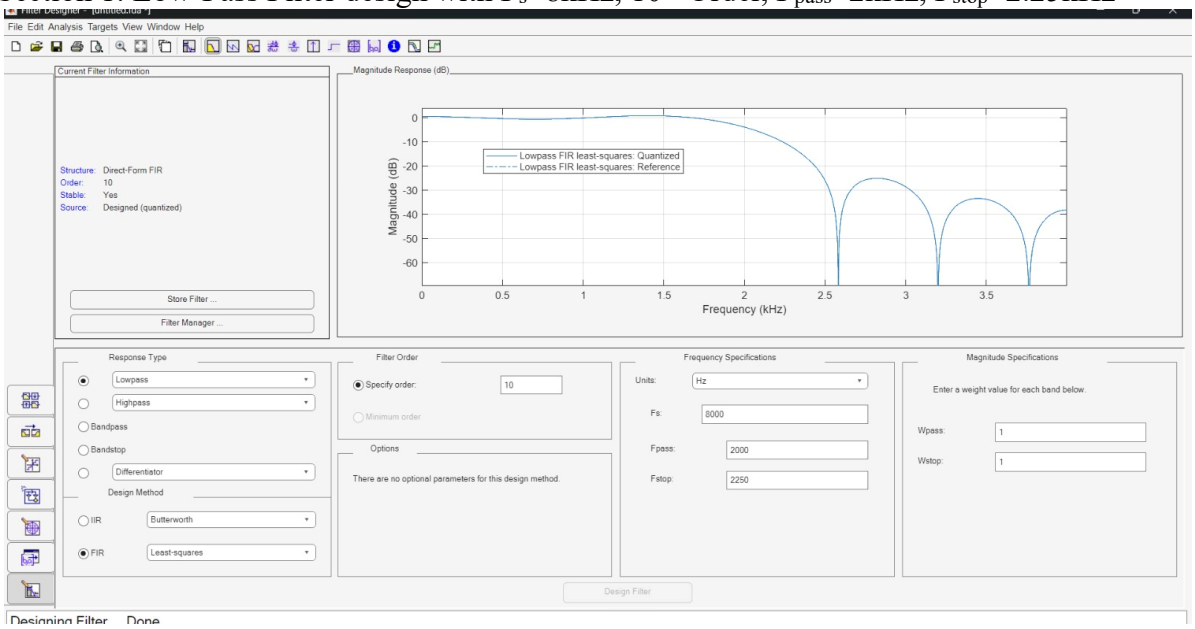


Figure 1: MATLAB Designer GUI

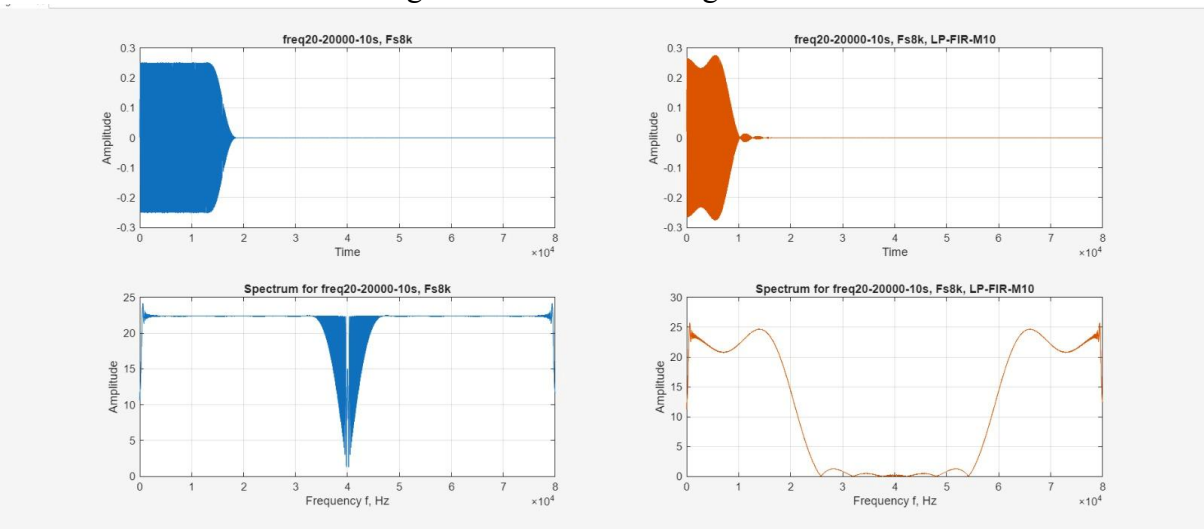


Figure 2: Input and Output Signal Graph

Section 2: High-Pass Filter design with F_s=8kHz, 10th Order, F_{pass}=2kHz, F_{stop}=2.25kHz

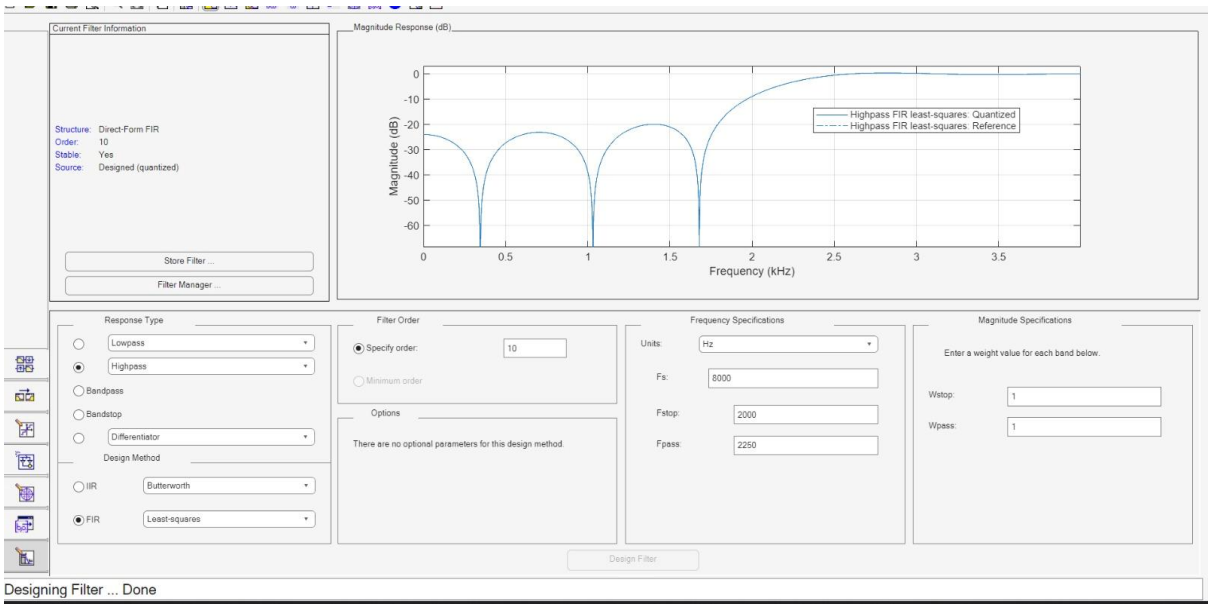


Figure 3: MATLAB Designer GUI

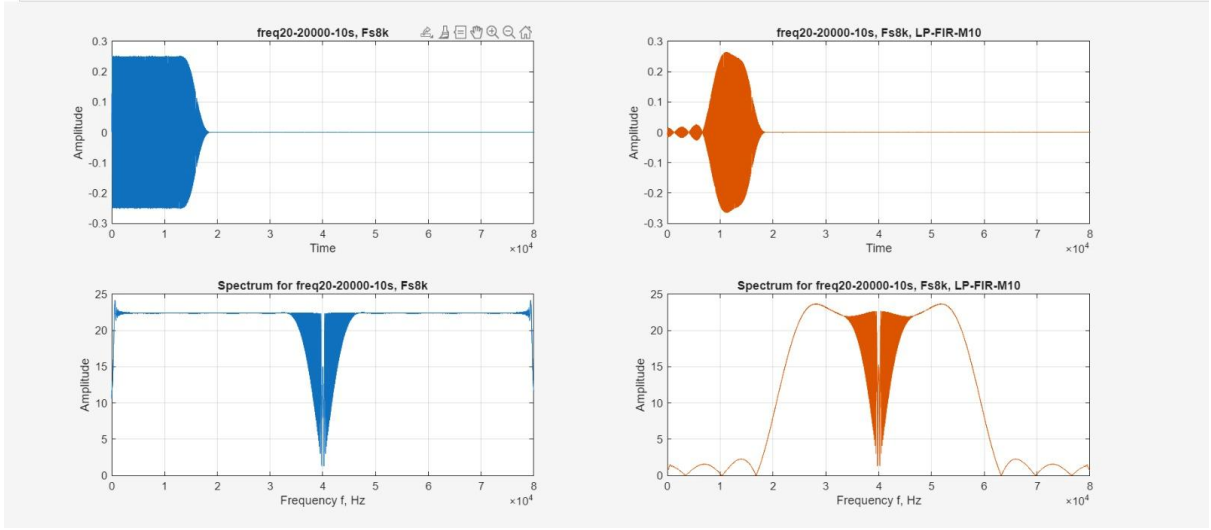


Figure 4: Input and Output Signal Graph

Section 3: High-Pass Filter design with F_s=8kHz, 10th Order, F_{pass}=1.25kHz, F_{stop}=1kHz

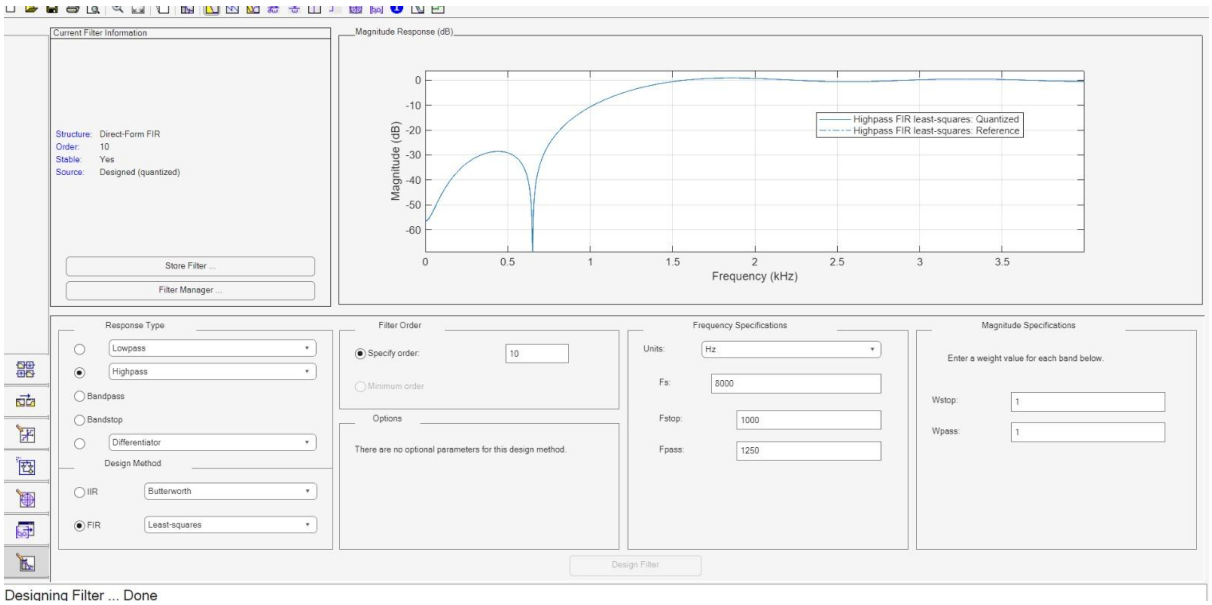


Figure 5: MATLAB Designer GUI

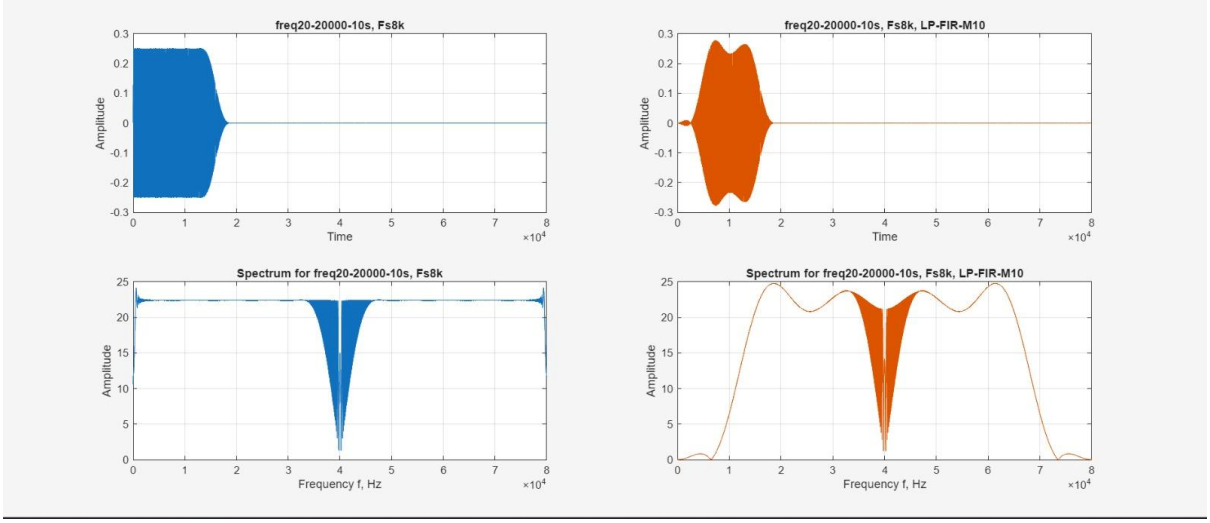


Figure 6: Input and Output Signal Graph

Section 4: High-Pass Filter design with F_s=8kHz, 20th Order, F_{pass}=1.25kHz, F_{stop}=1kHz

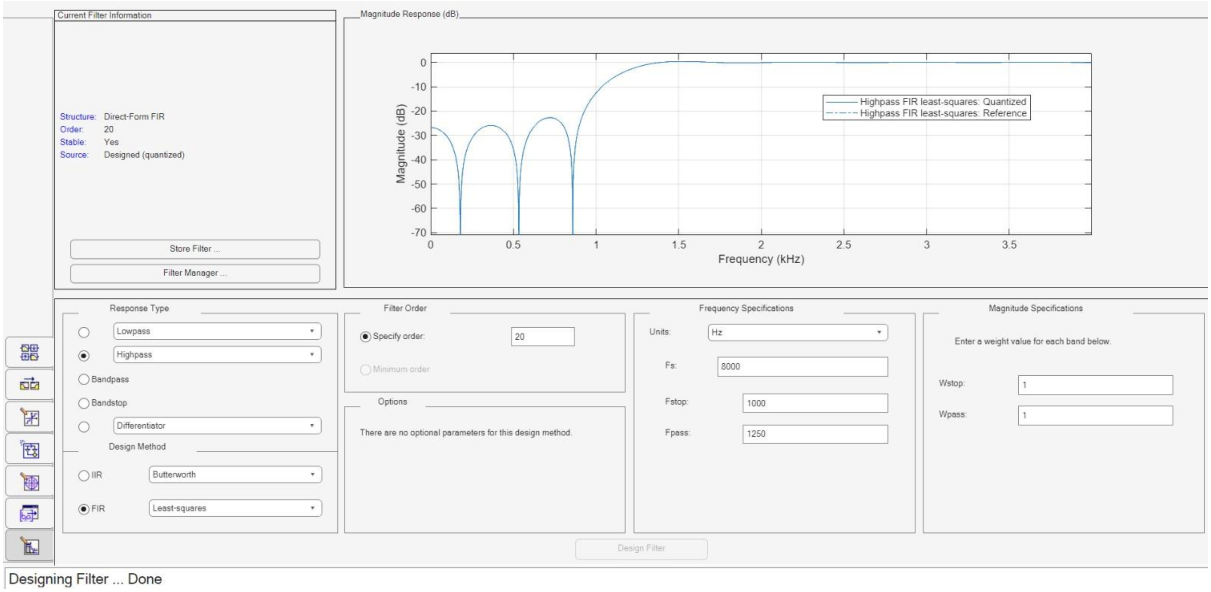


Figure 7: MATLAB Designer GUI

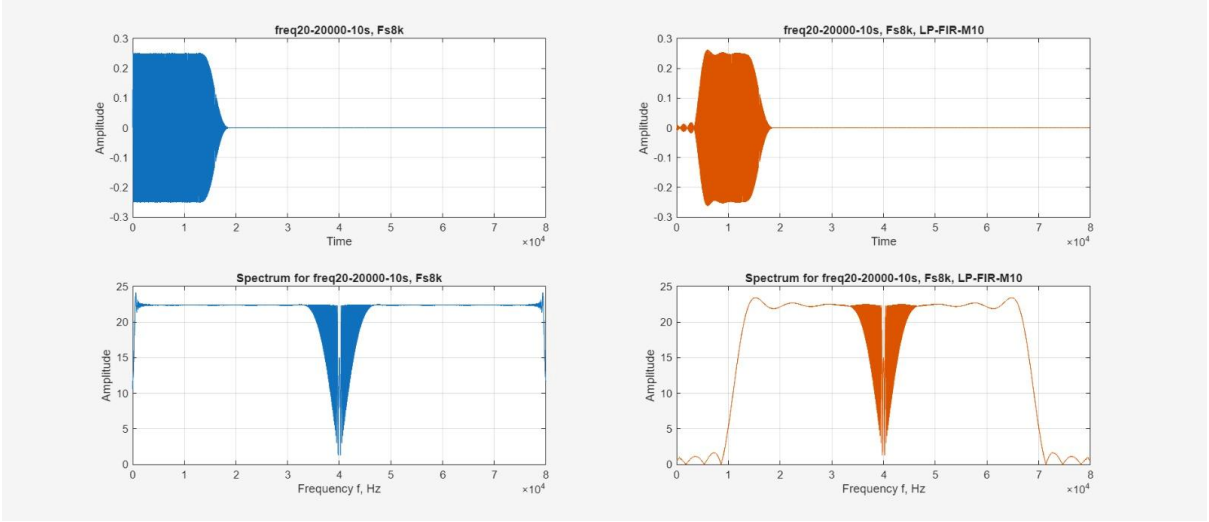


Figure 8: Input and Output Signal Graph

Section 5: Low-Pass Filter design with F_s=8kHz, 10th Order, F_{pass}=1.25kHz, F_{stop}=1kHz

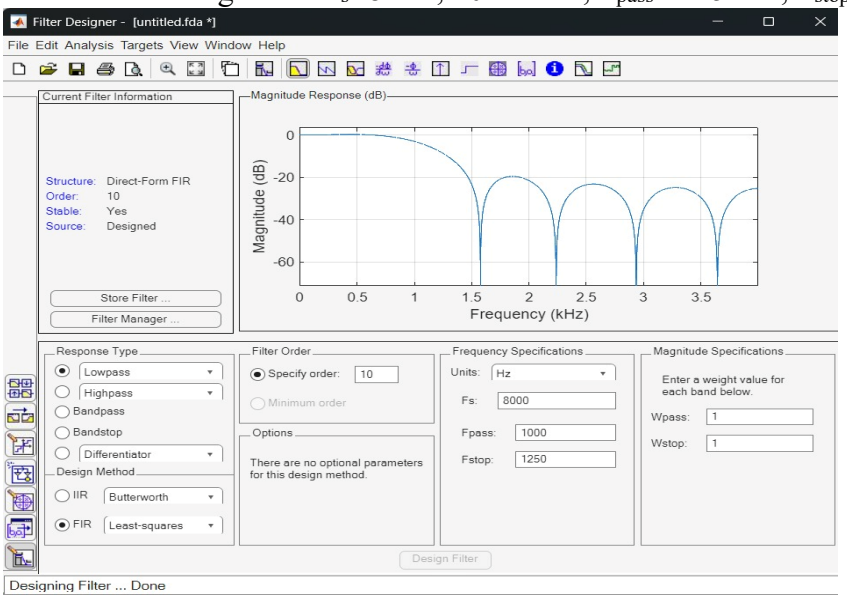


Figure 9: MATLAB Designer GUI

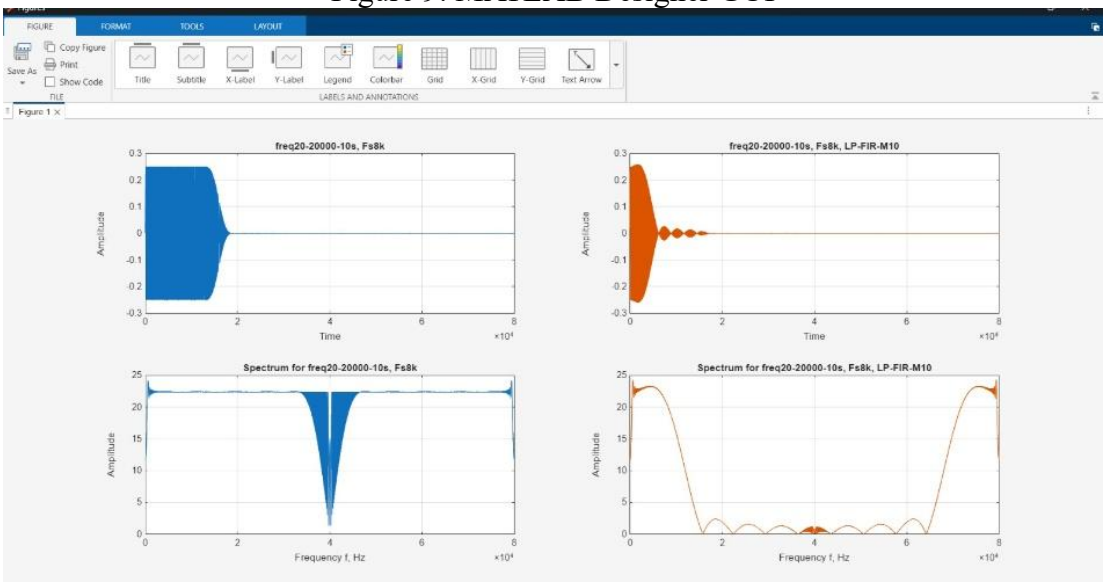


Figure 10: Input and Output Signal Graph

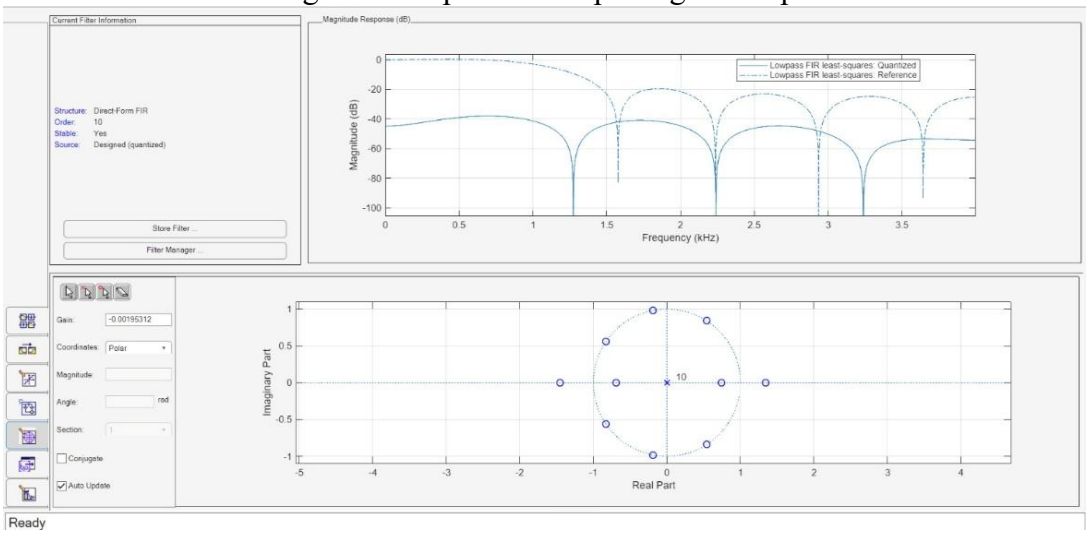


Figure 11: Filter analysis plot

Section 6: Low-Pass Filter design with F_s=8kHz, 20th Order, F_{pass}=1.25kHz, F_{stop}=1kHz

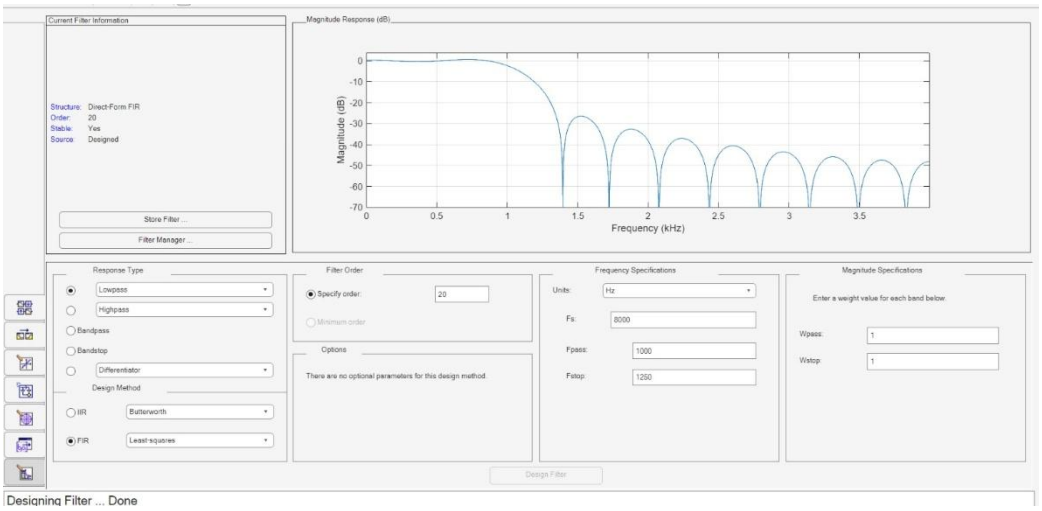


Figure 12: MATLAB Designer GUI

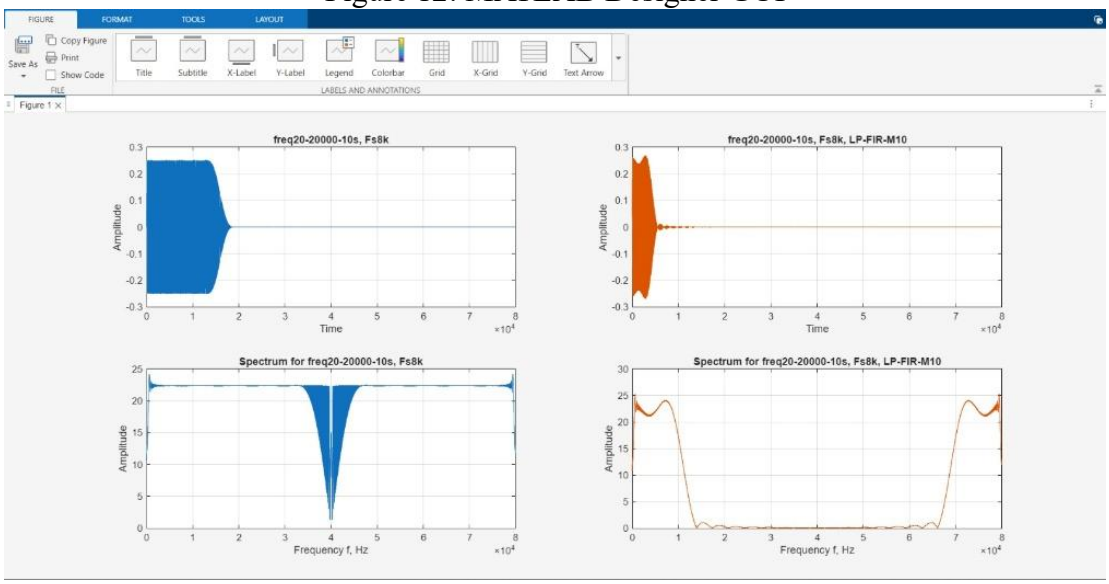


Figure 13: Input and Output Signal Graph

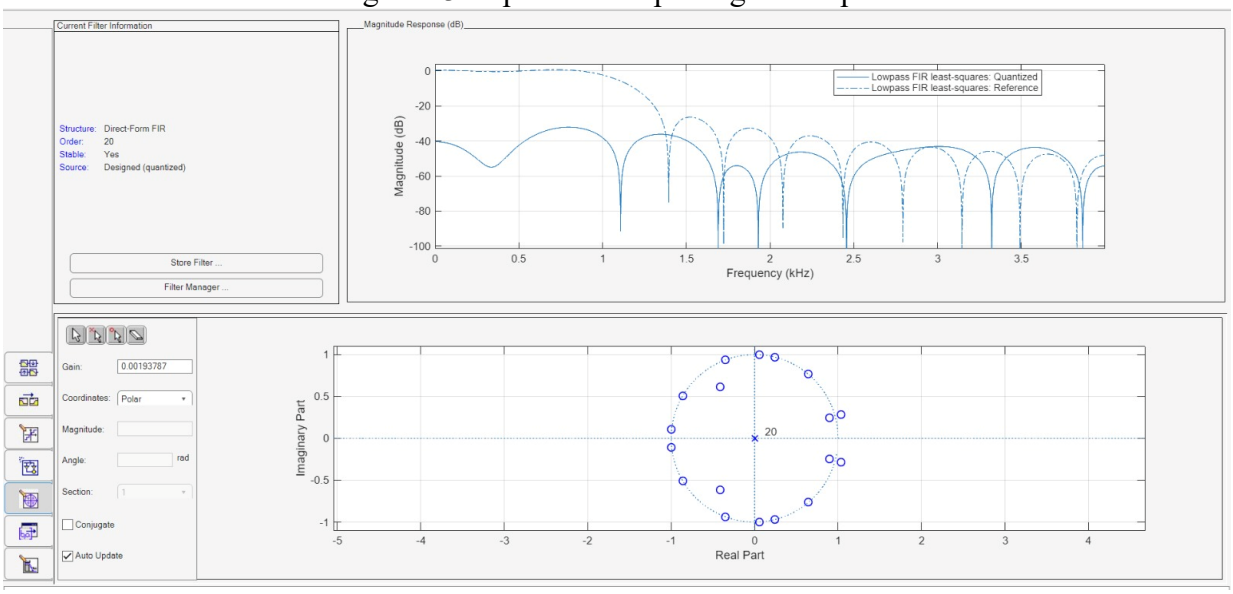


Figure 14: Filter analysis plot

Section 7: Low-Pass Filter design with F_s=8kHz, 30th Order, F_{pass}=1.25kHz, F_{stop}=1kHz

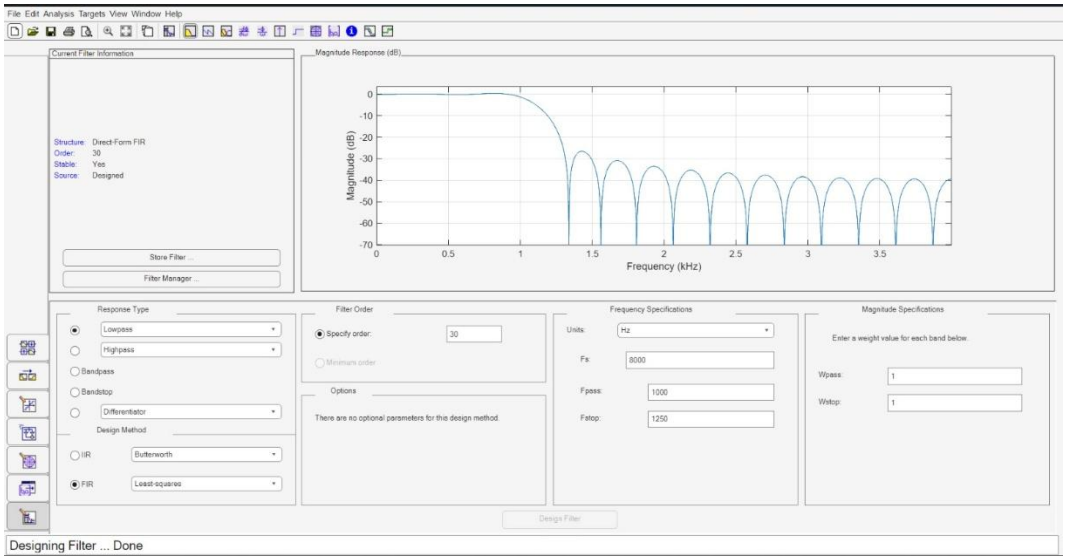


Figure 15: MATLAB Designer GUI

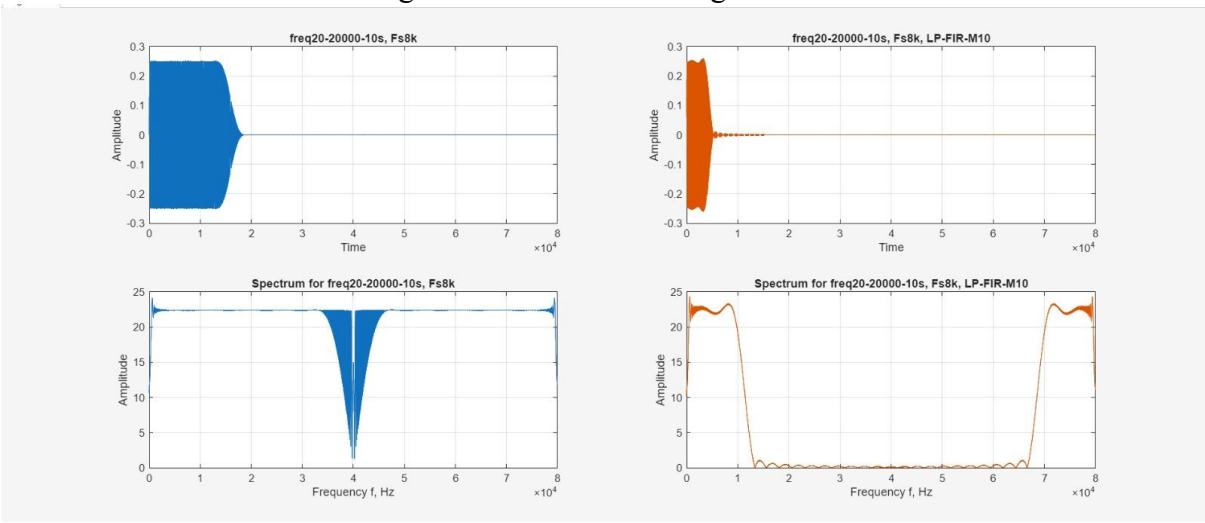


Figure 16: Input and Output Signal Graph

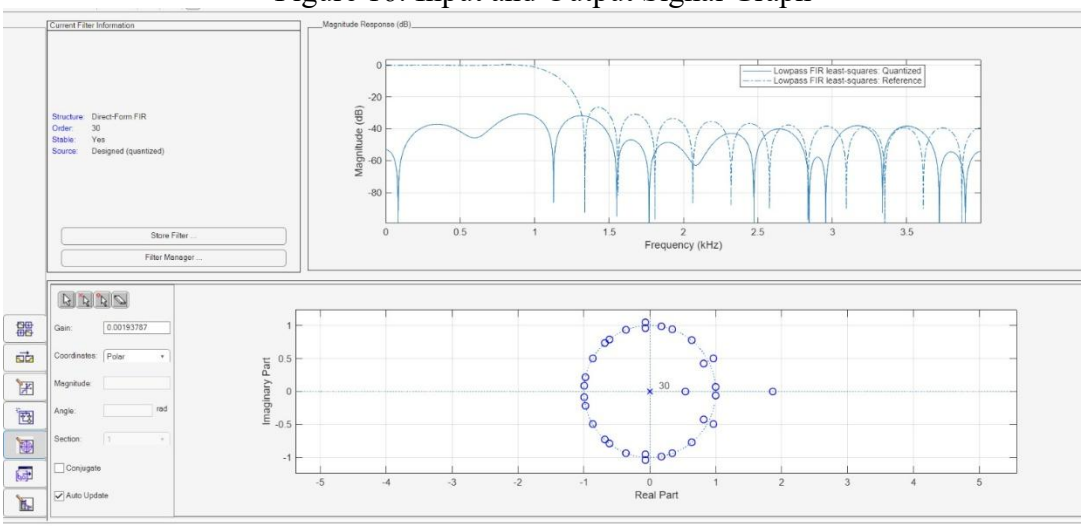


Figure 17: Filter analysis plot

Section 8: Low-Pass Filter design with F_s=16kHz, 10th Order, F_{pass}=1.25kHz, F_{stop}=1kHz

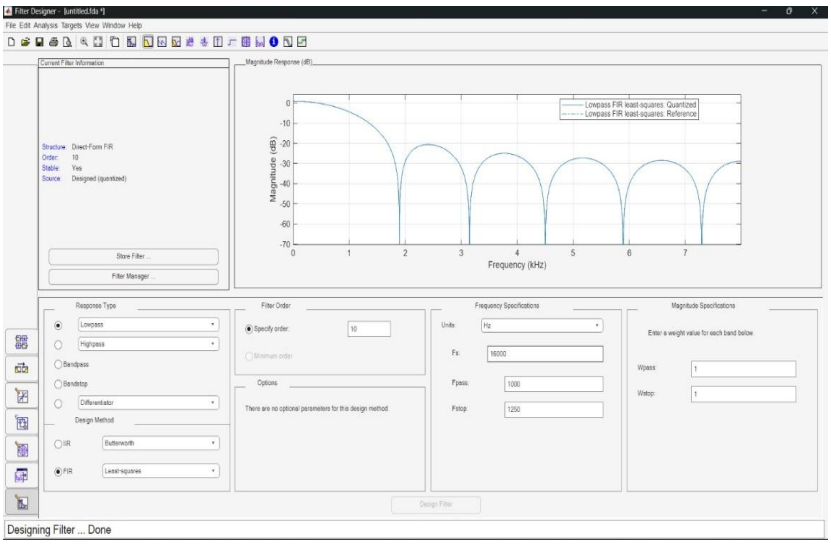


Figure 18: MATLAB Designer GUI

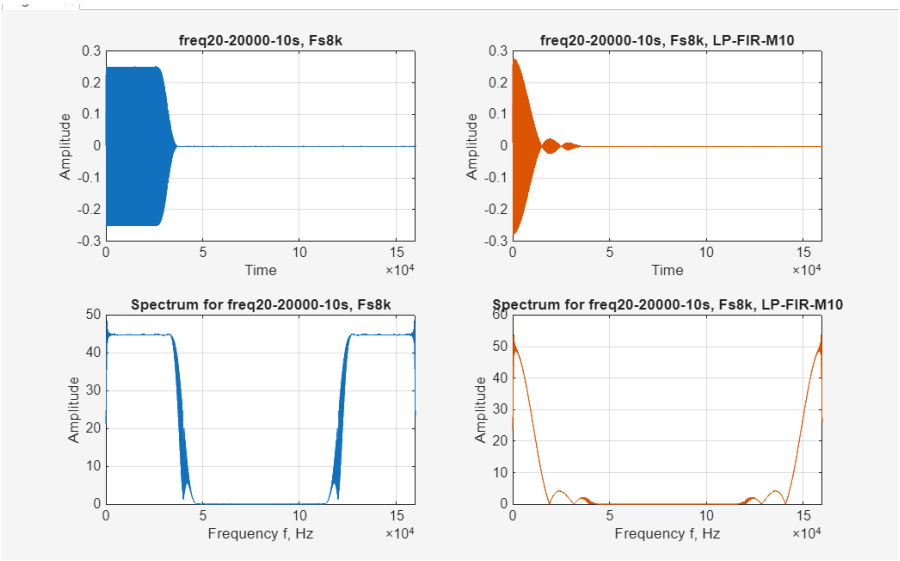


Figure 19: Input and Output Signal Graph

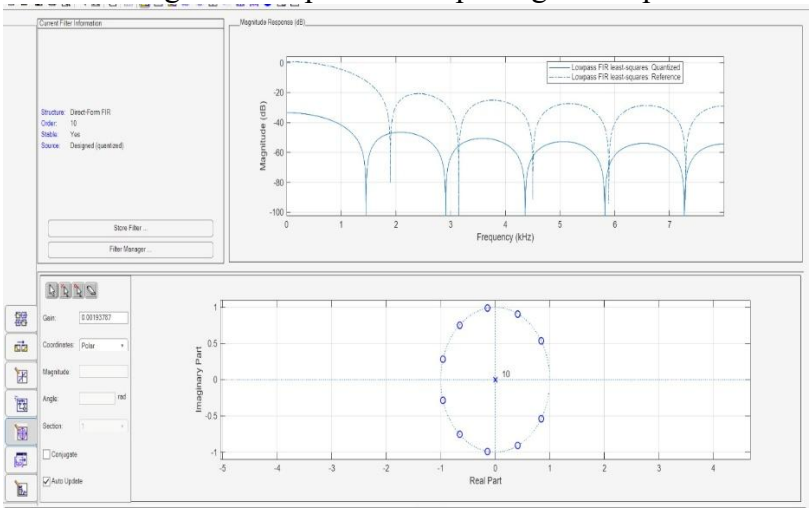


Figure 20: Filter analysis plot

Section 9: High-Pass Filter design with F_s=16kHz, 10th Order, F_{pass}=1.25kHz, F_{stop}=1kHz

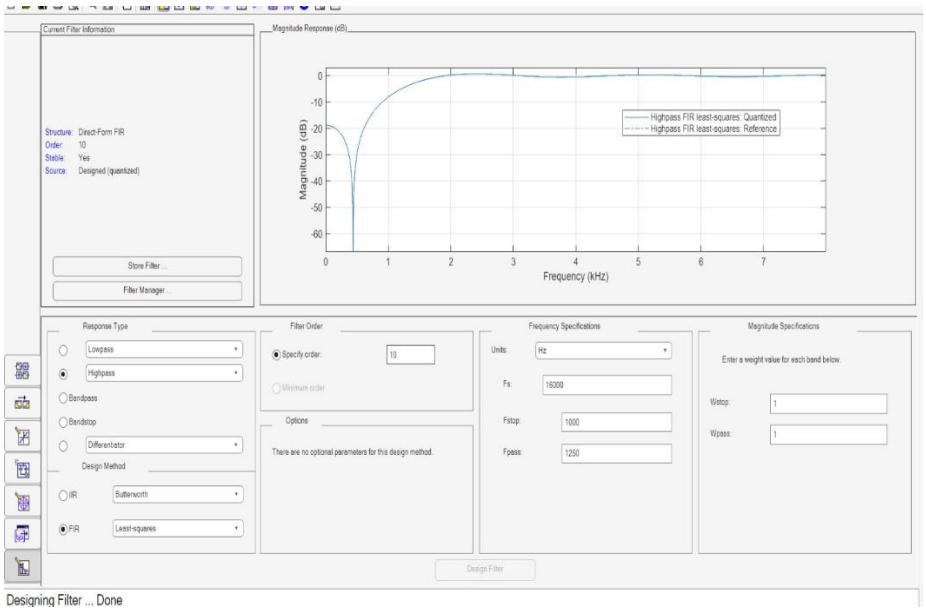


Figure 21: MATLAB Designer GUI

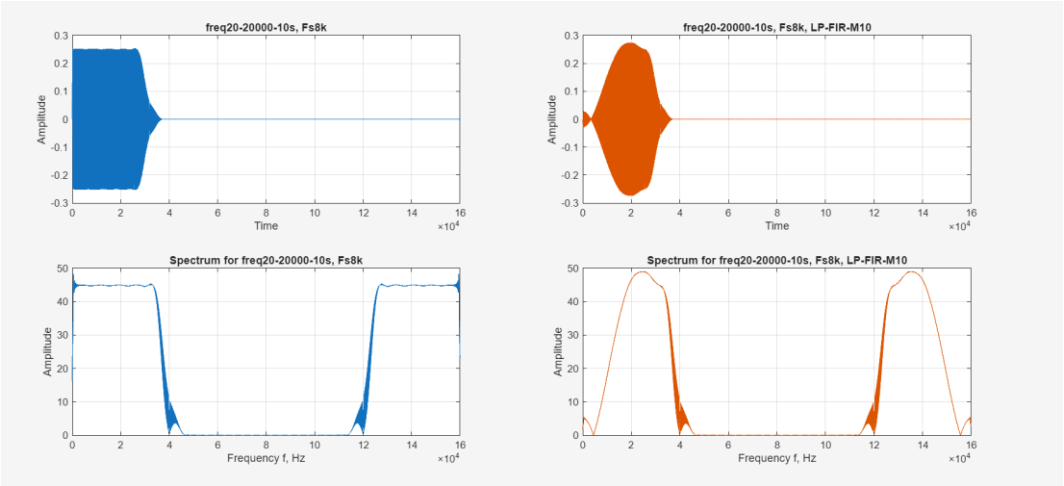


Figure 22: Input and Output Signal Graph

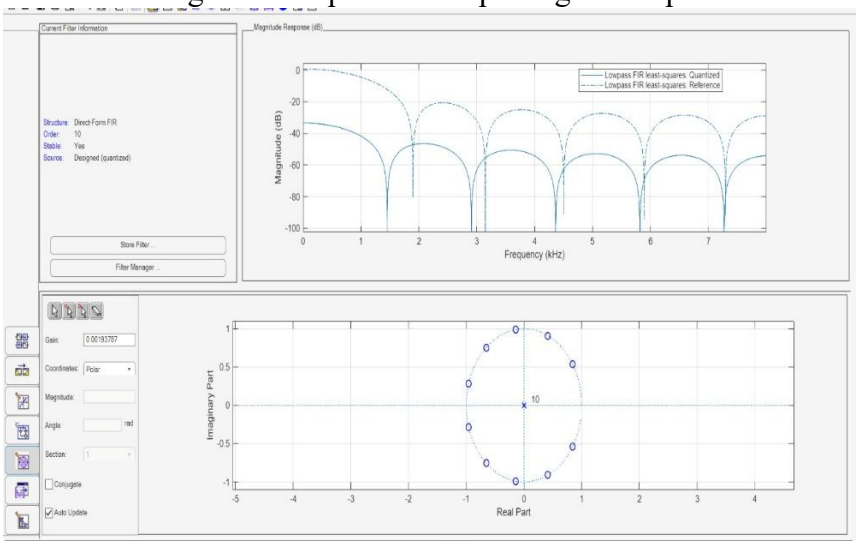


Figure 23: Input and Output Signal Graph

Discussions

On the general trend, the design of High and Low pass filter according to their respective output is being represented clearly in Figure 1 and 2 where the lower-frequency content is supressed above the stopband, indicating the presence of low-pass filtering pattern which reduces high-frequency component when attenuates a little on the low frequency component. On the contrary, for Figure 3 and 4, the characteristics are inverted due to the nature of the filter being a High Pass filter. The lower frequency spectrum is attenuated when the high frequency is preserved. Lastly, the comparison between Figure 2 and 4 shows how amendments regarding the filter type will affect the frequency behaviour in a sense that the are inversion of one another. This phenomenon shows that the changing of filter type from Low-pass to High—pass will result in a reverse of the frequency selective behaviour despite all other conditions remain the same.

The FIR filter can be computed using the following equation (MathWorks, n.d.).

$$equation\ 1: y[n] = \sum_{k=0}^{M} h[k]x[n-k]$$

When M is defined as the filter order, 10, 20, 30 in this case, and h[k] are the coefficients for FIR. In deduction, the transfer function of the FIR filter can be defined as

$$equation\ 2: H(z) = \sum_{k=0}^{M} h[k]\, z^{-k}$$

When the second equation is a Z-transform of the first equation. This follows the objective where a low-pass and high-pass filter is designed and verified by the observation in regards to the output when the filter type and M is changed in the design. The characteristic of the FIR filter is significant in every figure shown above, where the output slowly oscillates to zero, the point of stability. This shows that the design does not oscillates till infinity showing an infinite impulse response. On the contrary it slowly oscillates to zero indicating the finite characteristics of the FIR filter system. A characteristic from figure 20 and figure 21 illustrate how the FIR filter behaves. For figure 21, the low pass filter, the outputs stay close to 0dB during the pass band and increases after the edge of the passband. This indicates that it only let the frequency that is lower than the band to pass through. On the other hand, the high pass filters the low frequency region attenuates and approaches 0dB at the high frequency region.

The design of High and Low pass filter according to their respective output is being represented clearly in Figure 1 and 2 where the lower-frequency content is supressed above the stopband, indicating the presence of low-pass filtering pattern which reduces high-frequency component

when attenuates a little on the low frequency component. On the contrary, for Figure 3 and 4, the characteristics are inverted due to the nature of the filter being a High Pass filter. The lower frequency spectrum is attenuated when the high frequency is preserved.

By comparing section 3 and 4, both filters are FIR-High Pass filter with the same condition of High-Pass Filter design with Fs=8kHz, Fpass=2kHz, Fstop=2.25kHz, distinguished only by the order number (10th for Section 3 and 20th for Section 4) the presence of *h[k]* indicates that the higher the order number M, the higher the coefficient of FIR. This shows that the filter will have more liberty in design. This is shown by sharper high-pass response for figure 7 than figure 5, implying more zeros. Similarly, by comparing Figure 6 and 8, Figure 8 has a smoother change in signal indicating attenuation at t=1 in comparison to the sharp and drastic rise and fall at t=1 for figure 6. Such show that Figure 8, the 20th order filter has a stronger attenuation of unwanted signals.

In reference to the filter order, Figure 9 to 11 are a order 10 filter, when Figure 12 to 14 is a type of order 20 filter and figure 15 to 17 is a type of 0rder 30 filter. According to *equation 1*, when the order number M increases, the transition band becomes sharper, hence the unwanted band is rejected more distinctively. Thus, the FIR characteristics where higher order will result in more poles and zeros, hence more coefficient to achieve an ideal response. Figures 11,14 and 17 focus on the graphical representation of the different types of order of Low-Pass filter with same specification. Figure 10(order 10th) has the smoother graph in comparison to Figure 15(30th Order) as the later show a much abrupt and sharper change at approximate t=1.5s. The reasoning behind such phenomenon is because the higher the order number the higher the coefficient, the higher the degree of freedom it has to force the frequency to be within the response region (MathWorks, n.d.). This will result in the filter being able to maintain a stable position without oscillating too much for a high order filter in comparison to a low order filter. Hence the transition from passband to stop band becomes narrow, resulting in the sharper cutoff.

In reference to figure 11, 14 and 17, it can be observed that the number of zeros increases as the order of the filter increases. The reasoning is as such following the equation.

$$H(z) = \frac{h[0]z^M + h[1]z^{M-1} + \cdots + h[M]}{Z^M}$$

This shows that the zeros will increase as the order number M increases. As a result the circular markers representing the zeros in the pole zero plot will increase. Such increment is important

and significant because they represent that the filer has more control of variables. This will allow a stronger set of rejection in the stop band and a narrower transition band as proven in the previous paragraph.

Focusing on section 8 and section 9, they share the same characteristic of Fs=16kHz, 10th Order, Fpass=1.25kHz, Fstop=1kHz when section 8 is Low-pass filter and section 9 is High-pass filter. Both filter exhibits a similar characteristics as the low F_s filter mentioned in the above mentioned paragraph. Such represents the nature of the filter being consistent and does not easily affected by the changing of F_s. F_s is the sampling frequency, the changing of this factor does not have an effect on the general comparison trend between higher F_s and lower F_s is because the basic characteristic is being determined by the filter type and filter order.

Comparing between two low pass filter of different sampling frequency, specifically section 5 and section 8, both share the same specifications other than different sampling frequency of 8kHz (section 5) and 16kHz(section 8). The difference in sampling frequency F_s has resulted in difference in percentage of cutoff region. The cutoff region for figure 19(section 8) is significantly smaller in proportion in comparison to figure 10 of section 5. Figure 19 cutoff the frequency before *t=1s* when figure 10 shows the cutoff region at approximately *t=1s*. Hence showing the earlier has a smaller cutoff region than later. The reasoning of such is because of the following equation.

$$w = 2\pi \frac{f}{F_s}$$

When *f* is the frequency, which is constant for both filters. When F_s increases, w, the Nyquist frequency increases, hence the cutoff region shift to a lower normalized frequency ranged. This causes the above-mentioned effect despite there is no change in cutoff frequency.

Lastly, a short comparison between section 3 and section 9. The nature of high pass filter attenuating the lower frequency when passing the higher frequency in respect of threshold frequency will result in despite the cutoff frequency remain constant, however the graph is being rescaled due to the presence of Nyquist Frequency. Section 3 has a F_s of 8Hz, hence a Nyquist frequency of 4kHz when section 9 has a F_s of 16kHz, hence a Nyquist frequency of

8kHz. This will result in the response in figure 22 to be more compressed horizontally than figure 6 despite no change in filter type and other filter specifications.

Conclusion

In conclusion, the experiment firstly fulfilled the main objective of successfully designing a FIR Low-pass filter using the MATLAB Filter Designer. Furthermore, several variables were calibrated, for example, the filter was redesigned to form a High-Pass filter. Amongst the Low-Pass filter, filter order, F_{pass} is changed to obtain different characteristics of the filter. Same is applied for the High-Pass filter where order 10 and order 20 of it is being designed and calibrated to obtain the trend.

Several trend can be observed. Firstly in respect to the high and low pass filter. Low pass filter has successfully displayed its characteristics of attenuation in high frequency components when preserving the low frequency components. The opposite has been observed in High pass filter where it attenuates low frequency and only allow high frequency spectrum to pass. This symbolizes that the design is working in perfect condition.

Secondly, in reference to the 10th and 20th high pass filter. The increment of filter order has reflected as an improvement in filtering performance. This is shown by the steeper change after passband. The 20th order High pass filter has proven such y producing a sharper drop and a stronger attenuation of unwanted signals. The further proves the successful design of the high pass filter.

The comparison between 10th 20th and 30th low pass filter can be observed that in low pass filter, the higher the filter order the narrower band transition from pass band to stop band. Sharper transitions are also observed. The reasoning to such phenomenon is because when the order increases, the filter has a higher number of filter coefficient. This will allow the filter to be able to reject unwanted frequencies when at the same time becomes more selective. Such characteristics reflect a greater flexibility of the FIR filter in shaping the optimum output. Furthermore, the display of such change and application of FIR shows that the experiment has achieved the objective of understanding and utilizing FIR filter by varying the filter order.

Lastly, the changing of sampling frequencies affects the position of the cutoff region. Despite thew nature of filtering characteristic remain the same. This is due to the nature of increment of sampling frequencies increases the value of Nyquist frequency, hence providing the cutoff frequencies remain constant, on comparison it would seem that cutoff frequencies occupy a smaller size.

In conclusion, the experiment is a success as it provided us with the capability to manipulate the filter type, filter order and filter sampling frequency to obtain the optimum and most suitable filter using MATLAB Filter Designer.

References

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National Instruments (n.d.) 'WindFIR_Filter'. Available at: https://www.ni.com/docs/en-US/bundle/labwindows-cvi/page/cvi/libref/cviwindfir_filter.htm (Accessed: 31 March 2026).